

Indigenous Chromatography Resins for Biopharmaceuticals

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Founding Director

Protpure Tech Pvt. Ltd.

- Introduce Protpure's Agarose-based Chromatography capabilities
- Discuss relevance for R&D and early process development
- Explore low-risk technical evolution, not supplier replacement

- Chromatography resins are critical by largely import-dependent
- Long-Lead time and limited flexibility during R&D iterations
- Increasing importance of optional indigenous materials for resilience

- Indian manufacturer focused on agarose-based chromatography resins
- Core expertise in bead synthesis, cross-linking and ligand chemistry
- Products developed for protein purification workflows

Size Exclusion Chromatography



Plain/ Standard/ Basic Agarose Resins

- 4% Agarose Resin
- 6% Agarose Resin

Activated/Crosslinked Agarose Resins

- 2% Activated Agarose Resin
- 4% Activated Agarose Resin
- 6% Activated Agarose Resin

Ion Exchange Chromatography



SP Agarose Resin

Sulfopropyl Agarose Resin
(Strong cation exchanger)

CM Agarose Resin

Carboxymethyl Agarose Resin
(Weak cation exchanger)

Q Agarose Resin

Quaternary amine Agarose Resin
(Strong anion exchanger)

DEAE Agarose Resin

Diethylaminoethyl Agarose Resin
(Weak anion exchanger)

Affinity Chromatography



- Ni-NTA Agarose Resin
- Co-NTA Agarose Resin
- Zn-NTA Agarose Resin
- Cu-NTA Agarose Resin

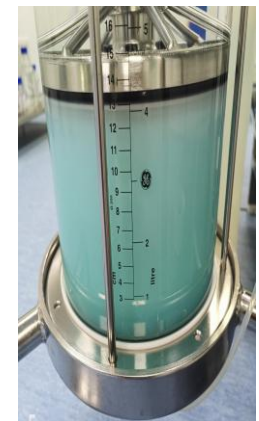
Protpure Facility Snapshot

Attribute	Details
Location	Anand, Gujarat, India
Established	May 2023
Facility Type	Semi-automated manufacturing + R&D
Current Capacity	600 L/month production of agarose-based resins
Product	Used in GMP facilities; repeat orders from Indian biopharma companies
Validation	
Expansion Plans	Active planning for large-scale production plant
Team Strength	8-10 employees; founder-led innovation
Funding Status	Bootstrapped

“Protpure is building India’s next-generation resin platform — combining technical depth, reproducibility, and customization for protein, peptide, and small molecule purification.”

Protpure Vision

- Democratize access to high-quality agarose resins for Indian biopharma
- Reduce dependency on imports and foreign exchange outflow
- Enable faster, cost-effective downstream processing with local innovation
- Build a resin platform that scales from R&D to commercial manufacturing



- Ion Exchange Chromatography
 - Cation Exchanger – SP Agarose, CM Agarose
 - Anion Exchanger – Q Agarose, DEAE Agarose
- Affinity Chromatography
 - Ni-NTA Agarose (His-tagged proteins)
- SEC & Desalting
 - Plain and cross-linked agarose matrices

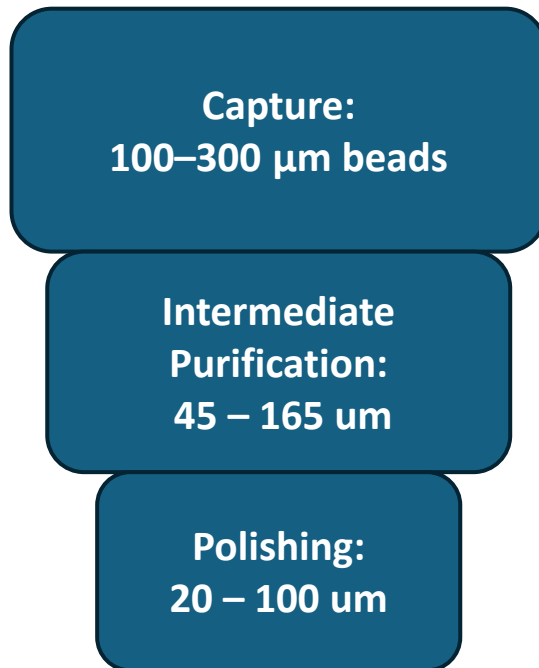


- Core step in most biopharma purification processes
- Used for capture, intermediate and polishing stages
- High impact on yield, purity and robustness

Our design focus:

- Controlled bead size distribution
- Stable ligand attachment
- Reproducible batch characteristics

Bead Size vs. Resolution: Why It Matters ?



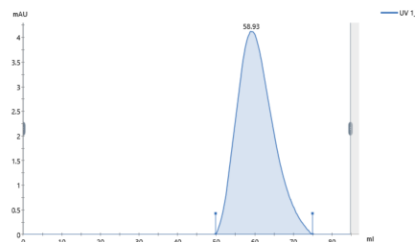
Purpose	Resolution	Typical Use Case
Bind and isolate large proteins or complexes from crude feedstock	★ Low	Initial step for mAbs, plasma proteins, vaccines
Remove bulk impurities, improve purity	★★ Moderate	Recombinant proteins, enzymes, hormones
Achieve high-resolution separation of closely related species	★★★ High	Isoform separation, aggregate removal, analytical SEC

Other parameters:

- Column dimension
- Bed volume
- Column pressure

SP Agarose – Strong Cation Exchanger

Parameter	SP Agarose	SP Agarose Precise	Industry Standard
BioProcess Resin	Yes	Yes	Yes
Ion Exchanger Type	Strong cation exchanger	Strong cation exchanger	Strong cation exchanger
Ionic Capacity	0.18–0.25 mmol (H ⁺)/mL resin	0.18–0.25 mmol (H ⁺)/mL resin	0.18–0.25 mmol (H ⁺)/mL resin
Ligand	Sulphopropyl	Sulphopropyl	Sulphopropyl
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	60–100 µm	45–165 µm
Particle Size (d50V)	~90 µm	~70 µm	90 µm
Dynamic Binding Capacity	≥150 mg Lysozyme/mL resin	≥150 mg Lysozyme/mL resin	≥150 mg Lysozyme/mL resin
Pressure/Flow Specification	500–700 cm/h, 0.15 MPa, 20 cm bed height	300–500 cm/h, 0.15 MPa, 20 cm bed height	300–600 cm/h, 0.1 MPa, 10 cm bed height
pH Stability (CIP)	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol



Peak	Retention (min)	Area (mAU·min)	Area %	Asymmetry 10%	Ext coeff (mg ⁻¹ ·mL·cm ⁻¹)	HEIP (min)
Peak A	58.932	45.37	100.00	1.63		0.07463

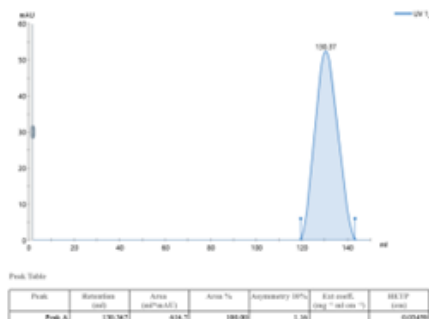
Column efficiency testing parameters	Protpure Result	Standard specifications
Peak asymmetry factor (A_s)	1.63	$0.80 < A_s < 1.8$
Reduced Plate height (h)	1.04	≤ 3

Applications

- **Purification of proteins, nucleic acids, and other biomolecules** with high selectivity.
- **Industrial-scale chromatography workflows** optimized for high-speed purification.
- **Reliable performance**, ensuring reproducible results across various purification needs.

Q Agarose – Strong Anion Exchanger

Parameter	Q Agarose	Q Agarose Precise	Industry Standard
BioProcess resin	Yes	Yes	Yes
Ion Exchanger Type	Strong anion exchanger	Strong anion exchanger	Strong anion exchanger
Ionic Capacity	0.18–0.25 mmol (Cl ⁻)/mL resin	0.18–0.25 mmol (Cl ⁻)/mL resin	0.18–0.25 mmol (Cl ⁻)/mL resin
Ligand	Quaternary Amine	Quaternary Amine	Quaternary Amine
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	60–100 µm	45–165 µm
Particle Size (d50V)	~96 µm	~70 µm	90 µm
Dynamic Binding Capacity	≥140 mg BSA/mL resin	≥140 mg BSA/mL resin	40–70 mg BSA/mL resin (Fast Flow) / ≥160 mg BSA/mL resin (XL)
Pressure/Flow Specification	500–700 cm/h, 0.15 MPa, 20 cm bed height	300–500 cm/h, 0.15 MPa, 20 cm bed height	300–600 cm/h, 0.1 MPa, 15 cm bed height
pH Stability (CIP)	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol



Column efficiency testing parameters	Protpure Result	Standard specifications
Peak asymmetry factor (A_s)	1.63	$0.80 < A_s < 1.8$
Reduced Plate height (h)	1.04	≤ 3

Applications

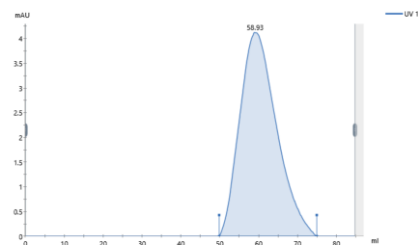
Purification of proteins, nucleic acids, and other biomolecules.

Suitable for large-scale manufacturing and research applications.

Reliable performance in high-speed chromatography workflows.

DEAE Agarose – Weak Anion Exchanger

Parameter	DEAE Agarose	DEAE Agarose Faster	Industry Standard
BioProcess Resin	Yes	Yes	Yes
Ion Exchanger Type	Weak anion exchanger	Weak anion exchanger	Weak anion exchanger
Ionic Capacity	0.12–0.15 mmol (Cl ⁻)/mL resin	0.12–0.15 mmol (Cl ⁻)/mL resin	0.88–1.08 mmol (Cl ⁻)/mL resin or /mg dry material
Ligand	Diethylaminoethyl	Diethylaminoethyl	Diethylaminoethyl
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	60–100 µm	45–165 µm
Particle Size (d50V)	~90 µm	~70 µm	90 µm
Dynamic Binding Capacity	≥90 mg BSA/mL resin	≥90 mg BSA/mL resin	40–100 mg BSA/mL resin or g/L bed vol
Pressure/Flow Specification	500–700 cm/h, 0.15 MPa, 20 cm bed height	300–500 cm/h, 0.15 MPa, 20 cm bed height	300–600 cm/h, 0.1 MPa, 10 cm bed height, 35 cm/h, 0.069 MPa (dry to wet bed)
pH Stability (CIP)	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol



Peak Table

Peak	Retention (min)	Area (mAU*min)	Area %	Asymmetry 10%	Ext. coeff. (mg ⁻¹ ·mL·min ⁻¹)	HEEP (cps)
Peak A	58.932	45.37	100.00	1.63		0.07963

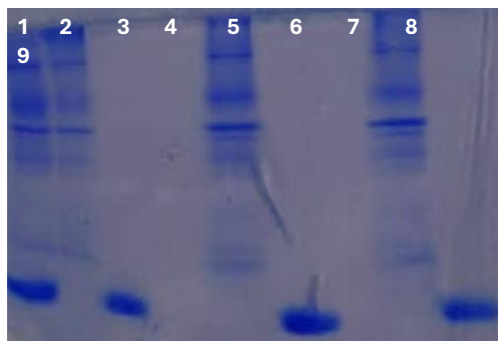
Column efficiency testing parameters	Protpure Result	Standard specifications
Peak asymmetry factor (A _s)	1.63	0.80 < A _s < 1.8
Reduced Plate height (h)	1.04	≤ 3

Applications

- **Purification of proteins, nucleic acids, and other biomolecules** with high selectivity.
- **Industrial-scale chromatography workflows** with optimized flow performance.
- **Reliable performance in high-speed purification processes**, ensuring reproducible results.

Ni-NTA Agarose – Affinity resins

Parameter	Ni-NTA Agarose	Ni-NTA Agarose Faster	Industry Standard
BioProcess Resin	Yes	Yes	Yes
Metal / Ligand	Nickel	Nickel	Nickel
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	100–200 µm	45–165 µm
Particle Size (d50V)	~90 µm	~150 µm	90 µm
Dynamic Binding Capacity	≥40 mg histidine tagged protein	≥20 mg histidine tagged protein	10 - 40 mg histidine tagged protein
Pressure/Flow Specification	500–700 cm/h, 0.15 MPa, 20 cm bed height	800–1000 cm/h, 0.15 MPa, 20 cm bed height	300–600 cm/h, 0.1 MPa, 10 cm bed height, 35 cm/h, 0.069 MPa (dry to wet bed)
pH Stability (CIP)	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol



Lanes:

- 1) Crude,
- 2) Flow trough (round 1),
- 3) Elute,
- 4) Blank,
- 5) Flow trough (round 2),
- 6) Elute,
- 7) Blank,
- 8) Flow trough (round 3),
- 9) Elute

Applications:

- For **recombinant protein purification**
- Construct screening
- Tool protein purification
- Benchmarking downstream behavior
- Primary an R&D and early development tool



- Ni-NTA agarose supplied commercially with repeat usage
- IEC resins evaluated in industry-relevant workflows
- Emphasis on reproducibility over aggressive customization

Our approach: Let evaluation data guide decisions

- Process development teams evaluating alternatives
- Organizations seeking India-based second sourcing
- Process development collaborations

Evaluation strategy

- Lab-scale or bench-scale evaluation under existing SOPs
- Side-by-Side comparison if desired
- Feedback driven iteration
- Outcome: Technical data only

Why This May be Useful for Your Team

- Additional technical optionality during development
- Faster access and local scientific support
- Insight into indigenous material performance

Long-term View

- Gradual scale-up of manufacturing capability
- Alignment toward GMP-complaint Production
- Goal: Long-term, reliable Indian Supplier

The Smart Choice for Protein Purification

√	High Purity & Superior Yield:	Engineered to deliver exceptional purity and high protein recovery, ensuring optimized purification efficiency.
√	Reproducibility for Consistent Results:	Designed for high reproducibility, enabling reliable batch-to-batch performance with minimal optimization.
√	Optimized High Flow Rate:	Faster Purification & Increased Throughput. Supports rapid purification workflows, improving throughput and productivity in bio-pharma applications.
√	Short Delivery Time & Enhanced Procurement Efficiency:	Accelerated supply chain ensures timely availability for seamless operations.
√	Made in India for Reliability & Accessibility:	Locally manufactured, providing high-quality chromatography solutions while reducing reliance on imported alternatives.
√	Innovative & Cost-Effective:	Advancing Purification Technology Without Compromise at a competitive cost.

Strategic Opportunity for Indian Biopharma

“For biopharma teams currently using non-agarose resins under moderate to low pressure, Protpure’s agarose-based resins offer a compatible, high-resolution alternative — with GMP familiarity, reproducible performance, and scalable throughput.”

- We appreciate the opportunity for a technical discussion
- Open to feedback, critique and realistic assessment

Contact: Email: protpure@gmail.com, M: (+91) 9426596644



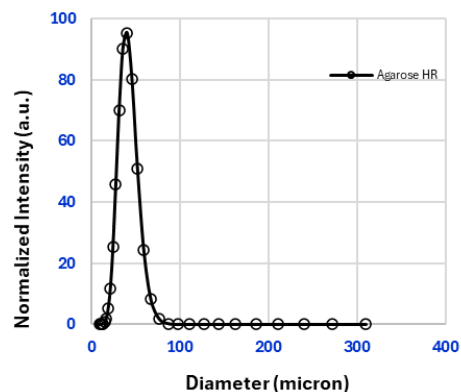
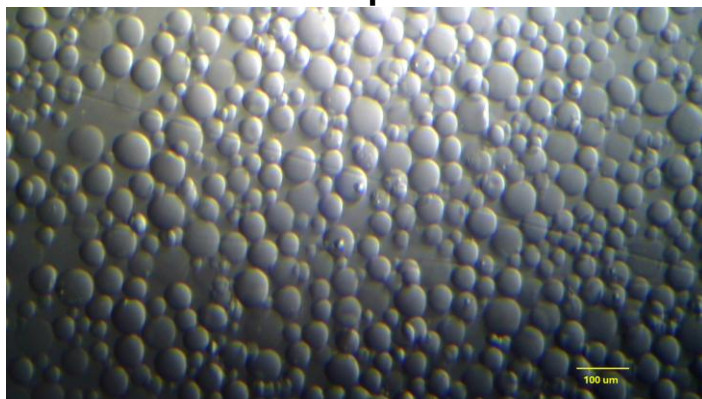
Thank You

Protpure : Linear Flow Velocity Highlights

Sr. No.	Designated Code	Characteristics	Linear Flow Velocity	Application Strength	Agarose Bead Mean size	Dynamic Binding Capacity
1	Agarose Faster	High Throughput Capability	Up to 1000 cm/hr	• Ideal for rapid processing workflow without compromising resolution for integrity. • Specially developed for Industrial requirements		
2	Agarose	Balanced Performance	Up to 700 cm/hr	• Harmonium balance of efficiency and separation clarity, suitable for most downstream processes.		
3	Agarose Precise	Precise Optimization	Up to 380 cm/hr	• Flow conditions favor high resolution separation for sensitive biomolecules and complex matrices		
4	Agarose HR	Gentle Elution	Up to 120 cm/hr	• Enable gentle handling of fragile proteins and minimizing shear, perfect for delicate purification steps		

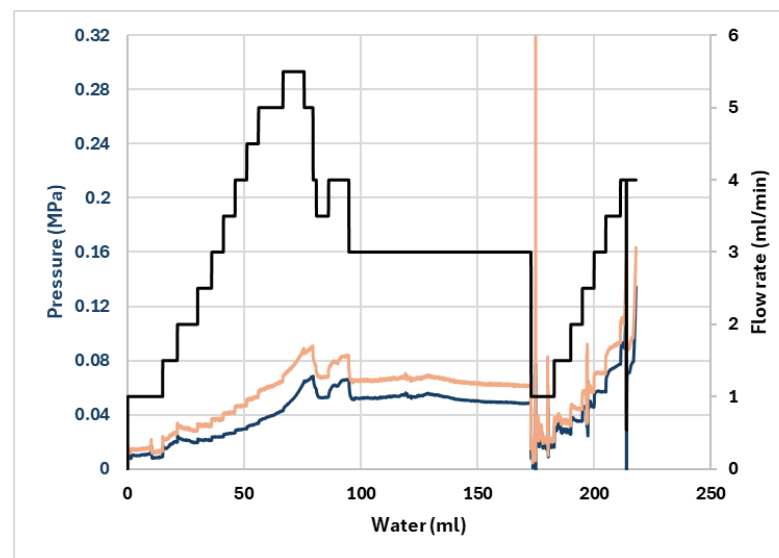
Agarose HR – Base matrix & crosslinked

Microscopic view



Mean size (µm)	Size range (µm)
40	15-75

Pressure / Flow Characteristics



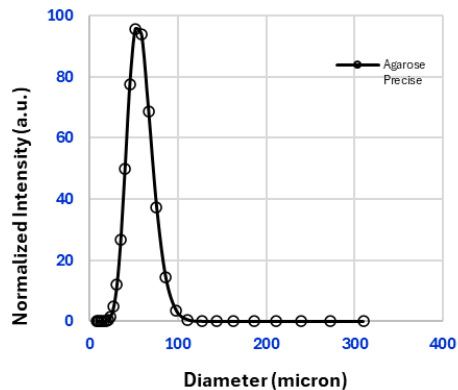
Column: MXK 26/20
Bed height: 20 cm
Bed volume: 106 cm

Agarose HR:
Linear Flow Velocity : up to 120
cm/hr,
Prec Pressure < 0.1 Mpa

- Agarose resins used for gentle elution with higher resolution.
- It can be used for the purification of small proteins and peptides.

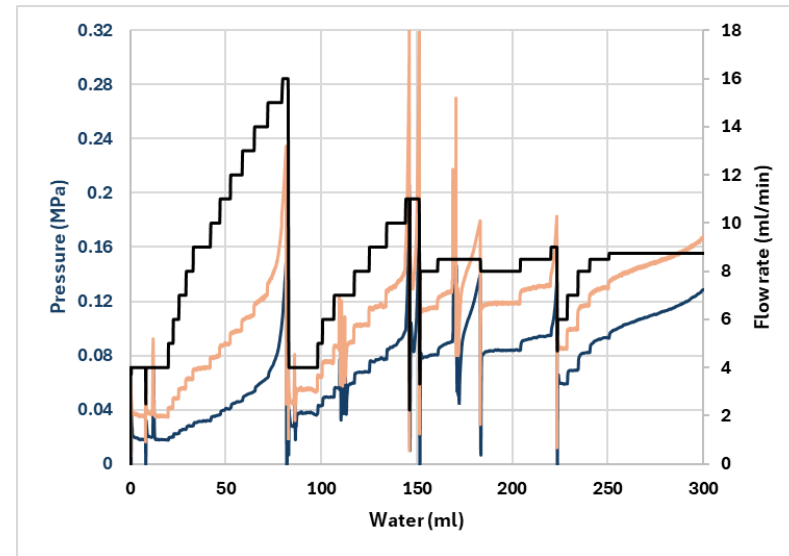
Agarose Precise – Base matrix & crosslinked

Microscopic view



Mean size (µm)	Size range (µm)
60	25-110

Pressure / Flow Characteristics



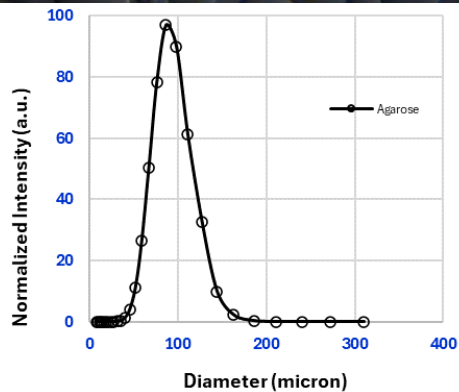
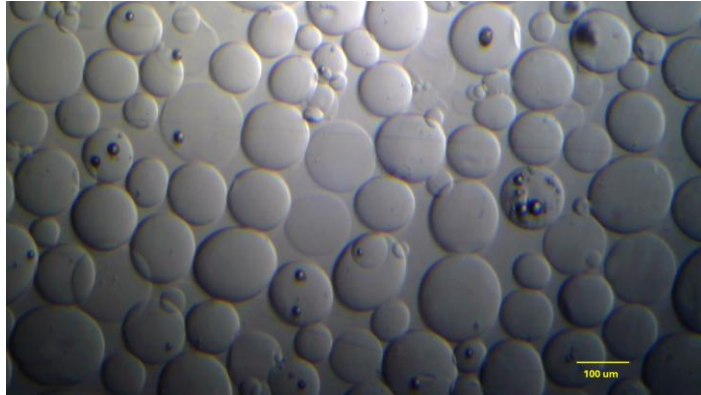
Column: MXK 26/20
Bed height: 20 cm
Bed volume: 106 cm

Agarose Precise:
Linear Flow Velocity : up to 380
cm/hr,
Prec Pressure < 0.12 Mpa

- Agarose resins used for precision optimization.
- It can be used for the purification of medium to small proteins.

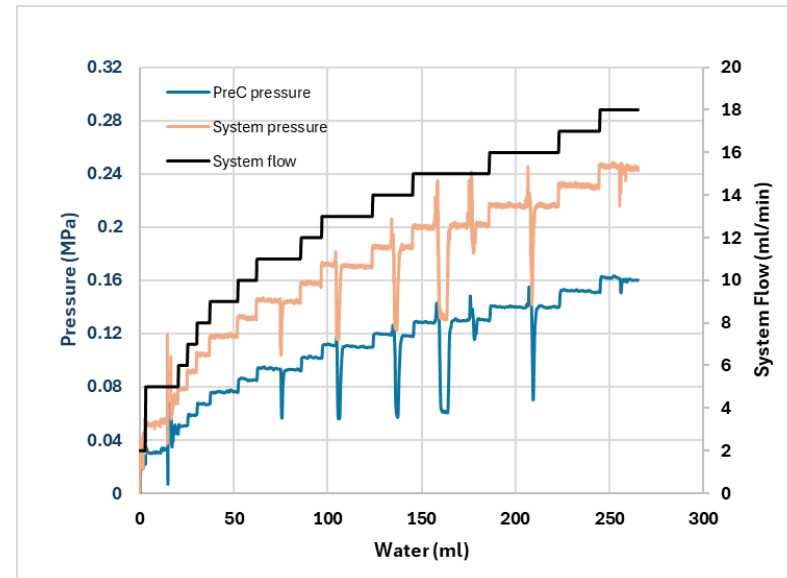
Agarose – Base matrix & crosslinked

Microscopic view



Mean size (um)	Size range (um)
96	45-165

Pressure / Flow Characteristics



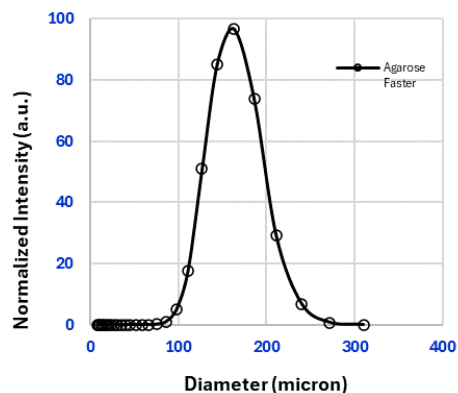
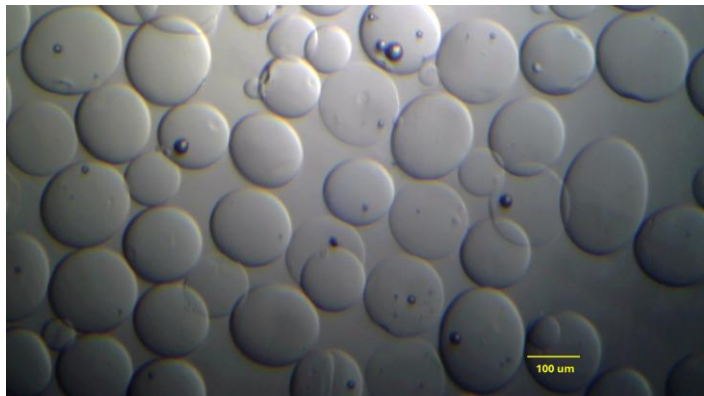
Column: MXK 26/20
Bed height: 20 cm
Bed volume: 106 cm

Agarose:
Linear Flow Velocity : up to 700 cm/hr,
Prec Pressure < 0.15 Mpa

- Agarose resins has balanced performance, i.e. relative high linear flow velocity with high DBC.
- It can be used for the purification of large to medium proteins, e.g. Recombinant proteins, partial purification of insulin etc.

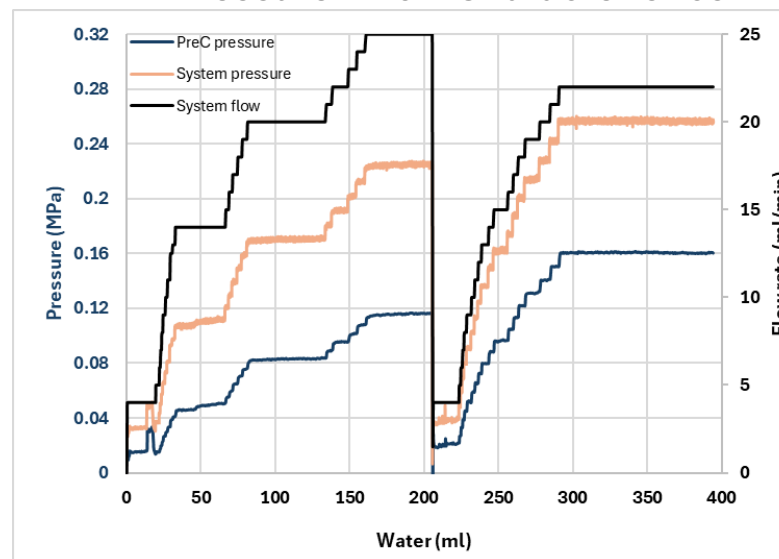
Agarose Faster – Base matrix & crosslinked

Microscopic view



Mean size (um)	Size range (um)
163	100-240

Pressure / Flow Characteristics



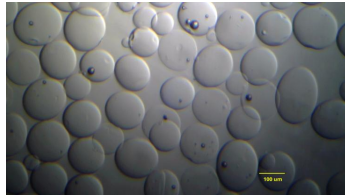
Column: MXK 26/20
Bed height: 20 cm
Bed volume: 106 cm

Agarose Faster
Linear Flow Velocity :
up to 1000 cm/hr,
Prec Pressure < 0.15 Mpa

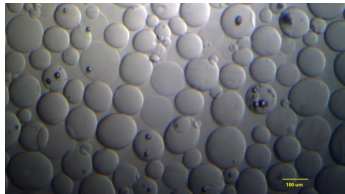
- Agarose Faster resins has high throughput capacity.
- It can be used for the purification of large proteins, e.g. Recombinant proteins, partial purification of insulin etc.

This variant is specially developed for our industrial collaborator.

Protpure Products



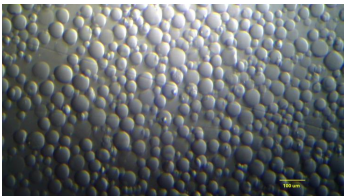
Agarose Faster



Agarose



Agarose Precise



Agarose HR

Ion Exchange Chromatography	• SP Agarose – Strong Cation Exchanger • Q Agarose – Strong Anion Exchanger • DEAE Agarose – Weak Anion Exchanger
Affinity Chromatography	• Ni-NTA Agarose • Co-NTA Agarose • Zn-NTA Agarose • Cu-NTA Agarose
Hydrophobic Interaction Chromatography	• Phenyl Agarose
Size Exclusion Chromatography	• Activated/Cross-linked Agarose resins

Agarose Variant	Mean Size	Linear Flow Velocity
Agarose Faster	163 µm	~ 1000 cm/hr
Agarose	96 µm	~ 700 cm/hr
Agarose Precise	60 µm	~ 380 cm/hr
Agarose HR	40 µm	~ 120 cm/hr

Protpure Products



Affinity Chromatography	- Ni-NTA Agarose - Co-NTA Agarose - Zn-NTA Agarose - Cu-NTA Agarose
Ion Exchange Chromatography	- SP Agarose – Strong Cation Exchanger - Q Agarose – Strong Anion Exchanger - DEAE Agarose – Weak Anion Exchanger
Hydrophobic Interaction Chromatography	Phenyl Agarose
Size Exclusion Chromatography	Activated/Cross-linked Agarose resins

Agarose Variant	Mean Size	Linear Flow Velocity
Agarose Faster	163 um	~ 1000 cm/hr
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